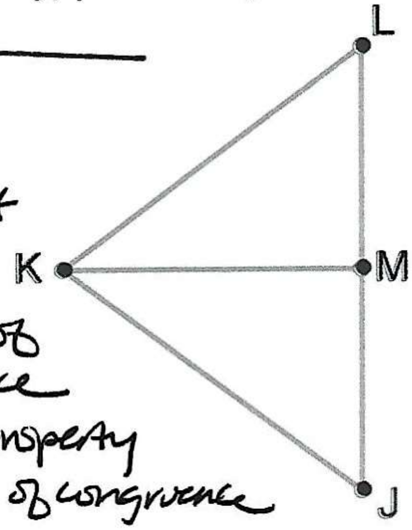
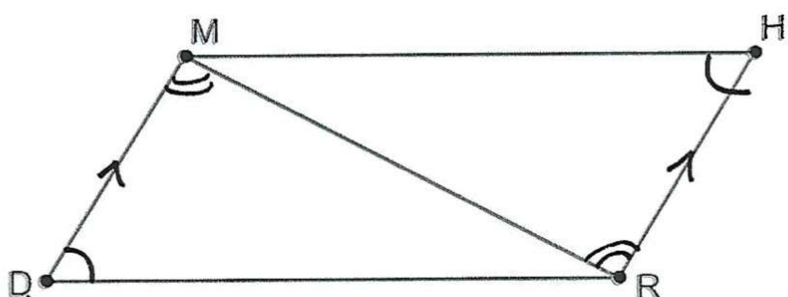
Unit 5: Worked Examples

1. Given $\overline{KM}$ bisects $\overline{LJ}$ and $\overline{KL} \cong \overline{KJ}$, prove $\Delta KMJ \cong \Delta KML$ using a two-column proof.



Statement	reason
1) $\overline{KL} \cong \overline{KJ}$	given
2) $\overline{KM}$ bisects $\overline{LJ}$	given
3) $LM = MJ$	definition of segment bisector
4) $\overline{LM} \cong \overline{MJ}$	definition of congruence
5) $KM = KM$	reflexive property
6) $\overline{KM} \cong \overline{KM}$	definition of congruence
7) $\Delta KMJ \cong \Delta KML$	SSS

1. Given $\overline{DM} \parallel \overline{RH}, \angle RDM \cong \angle RHM$, prove $\overline{DM} \cong \overline{RH}$ using a two-column proof.



Statement	reason
1) $\overline{DM} \parallel \overline{RH}$	given
2) $m\angle DMR = m\angle HRM$	alternate interior angles
3) $\angle DMR \cong \angle HRM$	definition of congruence
4) $\angle RDM \cong \angle RHM$	given
5) $\overline{MR} \cong \overline{MR}$	reflexive
6) $\Delta DMR \cong \Delta HRM$	AAS
7) $\overline{DM} \cong \overline{RH}$	CPCTC